MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE, EHU-39/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for single-pointer, self-balancing, potentiometer-type temperature indicator-type EHU-39/A ac white lighted.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environmental Resistant Removable Crimp Contacts, General Specification for
MIL-I-83262	Indicator, Temperature, Thermocouple, Self-Balancing Potentiometer

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6685

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General specification. Unless otherwise specified herein, the requirements of MIL-I-83262 shall apply. In case of conflict between MIL-I-83262 and this specification, this specification shall govern.

3.2 Certification. The contractor shall certify that the indicators furnished under this specification have passed the qualification tests in accordance with MIL-I-83262.

3.2.1 Power consumption. The indicator steady-state power consumption shall not exceed 7Va exclusive of the lighting circuit.

3.2.2 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of at least 1,000 hours at a 90 percent confidence level.

3.3 Design and construction. The indicator shall provide a counter-pointer presentation of temperature and shall incorporate a self-balancing potentiometer-type circuit for temperature measurement.

3.3.1 Power supply. The power supply shall be in accordance with MIL-I-83262 for ac powered indicators.

3.4 Performance. The indicator shall meet the following requirements:

a. Scale error not exceeding the tolerances as specified in table I

b. Friction error not exceeding 2.5°C for the counter and 5°C for the pointer

c. Position error not exceeding 2.5°C for the counter and 10°C for the pointer

d. Case leakage not exceeding 0.1 micron cubic feet per hour

e. Warmup and pointer travel from 100° to 1,000°C within 6 seconds at room temperature and 10 seconds at -55° and +100°C

f. Response to input signal changes from 500° to 850°C in not more than 2 seconds at room temperature and 2.5 seconds at -55° and +100°C with any combination of voltages and frequencies from 95 to 135V and from 320 to 480 Hz

g. When the signal is increased or decreased 0.08 mv, a change of 2° ±1°C indication shall be noted

h. Withstand temperatures ranging from -65° to +100°C

i. Withstand vibration as specified in Section 4

j. Withstand an alternate input of -3 and +52 mv for a period of 30 minutes at each condition for eight cycles with the ambient temperature at 50°C

k. Indication change not exceeding 2°C when a lead resistance change from 5 to 200 ohms is made

l. Indication steady-state change not exceeding 2°C when either the chromel or alumel conductor is grounded separately at any point to the ground side (low potential side) of the 115V 400 Hz power supply

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TABLE I. Test tolerances.

Test Points, °C	Scale Error at Room Temp., ±°C		Scale Error at Extreme Temp., ±°C		Scale Error During Reliability Test ±°C		Friction Error ±°C	
	Counter	Pointer	Counter	Pointer	Counter	Pointer	Counter	Pointer
200	5	15	10	20	10	20	2.5	5
300	5	15	10	20	10	20	2.5	5
400	5	15	10	20	10	20	2.5	5
500	5	15	10	20	10	20	2.5	5
600	5	15	10	20	10	20	2.5	5
700	5	15	10	20	10	20	2.5	5
800	5	15	10	20	10	20	2.5	5
900	5	15	10	20	10	20	2.5	5
1000	5	15	10	20	10	20	2.5	5
1100	5	15	10	20	10	20	2.5	5
1200	10	20	15	25	15	25	2.5	5
1300	10	20	15	25	15	25	2.5	5
1400	10	20	15	25	15	25	2.5	5

m. Withstand the effects of beat frequencies and stray ac signals in accordance with MIL-I-83262

n. Withstand pressures ranging from 30 inches Hg down to 0.309 inch Hg, (approximately sea level to 100,000 feet)

o. Withstand frequency variations ranging from 320 to 480 Hz, voltage variations from 95 to 135V steady-state and voltage surges to 195 ±5V

p. Withstand eight cycles of alternate immersion in tap water at 85° ±2°C and 5° ±2°C

q. Withstand a mechanical shock of 10g for 11 ±1 millisecond with no damage or malfunction of any component of the indicator

r. Exposure to salt sea atmosphere for a period of 48 hours.

3.5 Case. The indicator case shall be hermetically sealed and in accordance with MS33639 for the 2-inch diameter size. The case length shall be 7 ±0.25 inches exclusive of the electrical receptacle. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in lusterless black, Color No. 37038 of FED-STD-595. The remainder of the case shall be finished in black and shall be adequately protected to satisfactorily withstand the environmental test conditions specified herein. The hermetic seal shall not be accomplished by use of materials that are affected by the action of any atmosphere to which the case may be subjected. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than .03125 inch. The case shall be so designed that the internal mechanisms may be removed from the case, replaced, and the case resealed without the use of special tools, unless they are approved by the procuring activity.

3.6 Dial. The indicator dial shall be as shown on figure 1 and as otherwise specified herein. A minimum diameter of 1.75 inches shall be maintained across the outer ends of the scale graduations. The total scale arc shall be 240°

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±2°C. The scale shall be within 0.5 angular degree of precise linear spacing between adjacent scale graduations. The background of the dial shall be durable lusterless black consistent with integral lighting requirements.

3.7 Pointer. The pointer shall be in accordance with MS33585-8, except the tip of the pointer shall extend to the inner ends of the major scale graduations ±0.016 inch. The length of the lighted portion of the pointer shall be not less than 0.60 inch.

3.8 Counter. The indicator shall incorporate a four digit internal pinion mechanical counter as shown on figure 1. The presentation of the units digit shall be in even increments of two, i.e., 0, 2, 4, 6, 8, such that the display counts in increments of 2°C. The thousands digit shall have a plain black surface in the zero position so that only three digits are visible for indicated temperatures below 1,000°C. The numbers shall increase when the counter drums are rotated from top to bottom. The counter mechanism shall be directly coupled to the pointer drive mechanism.

3.9 Electrical receptacles. The indicator shall be provided with a hermetically sealed electrical receptacle in accordance with MIL-C-38999 type MS27476Y12B8P except pin A shall be made of alumel material and pin B shall be made of chromel material.

3.9.1 Wiring. The receptacle shall be wired as follows:

Pin	Electrical Connection
A	Alumel conductor
B	Chromel conductor
C	Case ground
D	Open
E	115V, 400 Hz input
F	115V, 400 Hz ground
G	5V AC or DC lighting input
H	5V AC or DC lighting ground
I	Warning switch
J	Warning switch

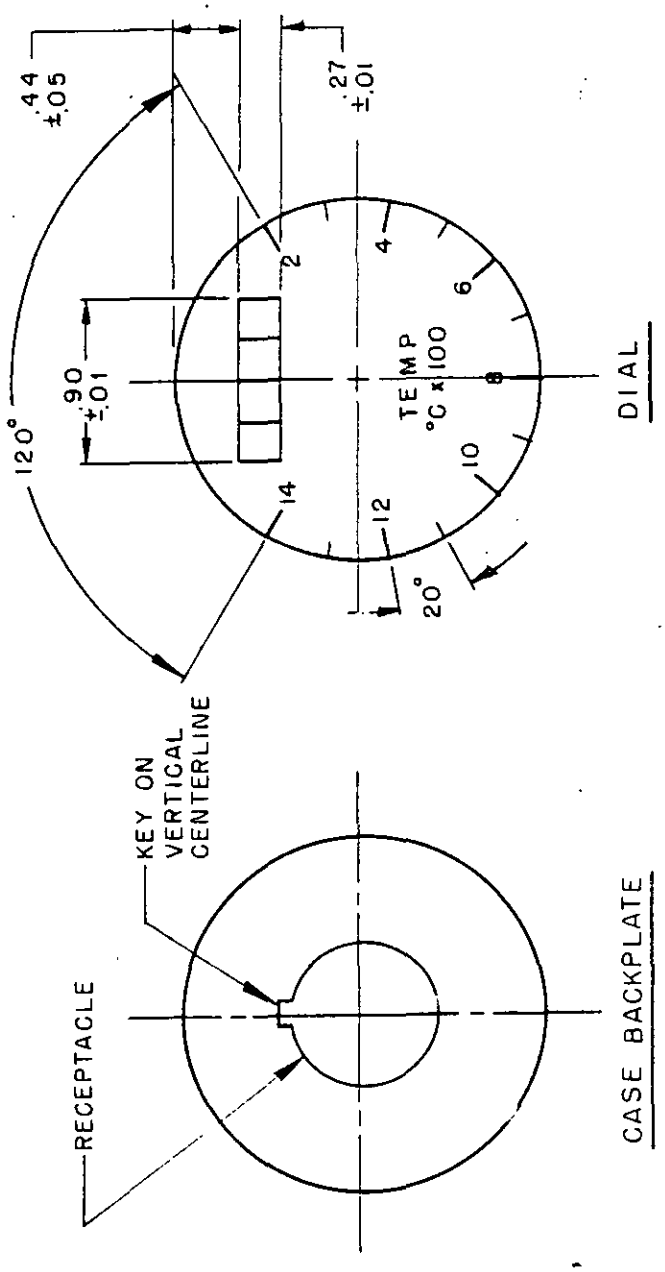
3.10 Warning switch. The indicator shall include a switch controlled by circuitry within the indicator. The switch shall be used to complete a circuit external to the indicator and shall be capable of carrying a maximum of 0.2 amperes at 28V DC. The switch point shall be internally adjustable over the temperature range of 950° to 1,050°C. The switch shall close with increasing temperature within the tolerances specified in table II.

TABLE II. Accuracy of switch point.

Ambient Temperature	Maximum Error
+25°C	±3°C
+100°C	±6°C
-54°C	±6°C

3.11 Marking dimensions and colors. The dial and pointer colors shall be in accordance with FED-STD-595. The numerals, markings, and letters shall be in accordance with MS33558 and the following dimensions:

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NOTE:

LINEAR DIMENSIONS IN INCH, TOLERANCES ANGLES ±0° 30'.

FIGURE 1. Dial and case backplate.

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	Height or Length Inch ±0.005	Colors
Scale Numerals	0.156	37875
Counter Numerals	0.188	37875
Major Graduations	0.156	37875
Minor Graduations	0.125	37875
Lettering TEMP °C x 100	0.094	37875
Background of Dial	-----	37038
Pointer	-----	37875

3.12 Excluded components. The design and fabrication of the indicator shall not include electromechanical choppers or plastic packaged integrated circuits.

3.13 Weight. The weight of the indicator shall not exceed 1.0 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, inspection and test

4.1.1 Sampling. Sampling shall be in accordance with MIL-I-83262.

4.1.2 Qualification inspection. Inspection for qualification shall be in accordance with MIL-I-83262 and as specified herein. Unless otherwise specified herein, the contractor shall perform the inspection and submit certification that the indicators conform to the requirements of this specification. A copy of the qualification test results shall be furnished to the Government.

4.1.2.1 Certification. Certification is authorized for this item under the following conditions:

a. Contractor's responsibilities

(1) The contractor shall be responsible for performance of all inspections and tests necessary to insure that the item submitted for Government acceptance is in accordance with all the requirements of the drawings and specifications listed in the contract.

(2) Prior to shipment of item, the contractor shall submit to the Government a certification of conformance based on either the contractor's inspection and tests, or those conducted by another qualified laboratory or inspection organization.

(3) The certification shall delineate all drawing and specification requirements being attested to, and shall be signed by a responsible officer of the contractor with authority to bind his principle.

(4) The certification shall contain the following statement: "I certify that the item described in this document has been inspected and tested and meets the following requirements. (Delineation of applicable requirements.)"

b. Government responsibilities. Item shall not be accepted until it has been properly identified with the certification. Government acceptance shall be made on the basis of contractor's certification of conformance; however, the Government reserves the right to independently inspect items for conformance to all contractual drawings and specification requirements before acceptance.

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4.1.3 Acceptance inspection. Acceptance inspection shall be in accordance with MIL-I-83262 except as otherwise specified herein.

4.1.4 Test methods. Unless otherwise specified herein, all test methods and procedures shall be in accordance with MIL-I-83262.

4.1.4.1 Power supply. Unless otherwise specified, the indicator shall be tested with 115V ±2V single-phase 400 ±5 Hz ac power and 4.5 ±0.5V ac or dc applied to the lighting circuit.

4.1.4.2 Scale error and friction. The friction error test shall be in accordance with MIL-I-83262 and shall be conducted at 600°, 700°, 800°, and 1,000°C.

4.1.4.3 Position error. This test shall be conducted at 600°, 800°, and 1,000°C test points.

4.1.4.4 Response time. Millivoltage signals and power shall be applied to provide a stable indication of 700°C. The millivoltage shall then be increased to produce an instantaneous change equivalent to 350°C. The time required for the indicator to fully respond to the 350°C change shall be noted and shall not exceed 2 seconds. This test shall be conducted using the following combinations of input voltage and frequencies: 115V 400 Hz, 95V 320 Hz, 95V 480 Hz, 135V 320 Hz and 135V 480 Hz. The test shall be repeated going from 1,050°C to 700°C. The same tolerance shall apply.

4.1.4.5 Sensitivity. The procedure shall be repeated at 600° and 900°C test points.

4.1.4.6 Thermocouple lead resistance effect. This test shall be conducted at 600°, 750°, and 900°C test points.

4.1.4.7 Grounding of thermocouple extension wires. Scale error shall be determined at the 600° and 900°C test points and the response time shall be determined in the 500° to 850°C range on increasing temperature.

4.1.4.8 Vibration. The indication shall be subjected to a random vibration test in accordance with MIL-STD-810, Method 514.2, equipment category b.2. The functional test level, W_O, shall be $0.018G^2/Hz$. The endurance test level, W_O, shall be $0.062\ G^2/Hz$ with test time per axis, T, equal to 2 hours.

4.1.4.9 Warning switch. Slowly vary the input thermocouple signal, both increasing and decreasing, to verify that with increasing temperature the switch closes within the tolerances specified in table II of the set switch point and that with decreasing temperature the switch opens at 2°C ±1°C below the actual switch closing point. Switch points of 950°, 980°, 1,010°, and 1,050°C shall be tested. The test shall be performed at each of the ambient temperatures as specified in table II with temperature stabilization of at least 3 hours with power off at each temperature. Power shall be applied a maximum of 1 minute total on-time prior to testing.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-I-83262.

5.2 Marking. In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

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5.2.1 Nomenclature. The nomenclature shall be as follows:

INDICATOR, TEMPERATURE
THERMOCOUPLE, SELF-BALANCING
POTENTIOMETER EHU-39/A

6. NOTES

6.1 All notes specified in MIL-I-83262 are applicable to this specification.

Custodian:
Air Force - 11

Reviewer activity:
Air Force - 99

Preparing activity:
Air Force - 11

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